

Notes: Angrist & Krueger (QJE, 1991)

Does Compulsory School Attendance Affect Schooling and Earnings?

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1 Big picture

- **Question.** Do compulsory school attendance laws increase schooling, and do the additional years of schooling induced by those laws raise earnings?
- **Core identification idea.** Quarter of birth interacts with school-entry rules and compulsory schooling laws. Children born early in the calendar year usually start school older and reach the legal dropout age earlier in their school careers. They can legally leave after completing less schooling than children born later in the year.
- **Natural experiment.** Date of birth is treated as as-good-as-random with respect to ability and family background. Quarter-of-birth variation therefore generates plausibly exogenous variation in educational attainment among people near the compulsory-schooling margin.
- **First-stage evidence.** Men born in the first quarter of the year have slightly less completed schooling and lower high-school graduation rates than men born later in the year. The effect is small in levels but precisely estimated because Census samples are very large.
- **Reduced-form evidence.** Men born in the first quarter also earn slightly less. The earnings pattern is weaker and noisier than the schooling pattern, but it moves in the direction predicted by the compulsory schooling story.
- **Main IV result.** Using quarter-of-birth dummies, and interactions with birth year and state of birth, as instruments for education, the two-stage least squares return to schooling is close to the OLS return. The paper reports estimates around 7–10 percent per year of schooling in the key middle-aged male samples.
- **Interpretation of the IV estimand.** The IV estimate is the return to schooling for people whose education is shifted by compulsory attendance laws—that is, students at the margin of dropping out around the legal school-leaving age.
- **Omitted-variable conclusion.** Since IV estimates are not lower than OLS estimates, the paper argues that conventional OLS estimates of returns to schooling are not strongly upward biased by omitted ability, at least for schooling near the compulsory-schooling margin.
- **Validation tests.** The paper checks that quarter of birth does not predict postsecondary education in the same way, that college graduates do not show quarter-of-birth earnings differences, and that the direct effect of compulsory schooling laws on enrollment is visible using state dropout-age variation.
- **Enduring importance.** The paper became a canonical example of instrumental variables in labor economics and a central reference point for later discussions of weak instruments, many instruments, LATE, and quarter-of-birth designs.

2 Paper structure and section-by-section roadmap

2.1 Introduction

The introduction motivates compulsory schooling as an institution present in developed countries but not well understood empirically. The authors point out that school-entry rules typically require children to have reached a certain age by a cutoff date before entering school. Compulsory attendance laws then require students to remain in school until a birthday, usually age 16 or 17. Together, these rules mean that students born at different times of the year face different legally required amounts of schooling.

Interpretation: The introduction sets up a clean empirical logic. Quarter of birth is not valuable because birthdays are economically meaningful by themselves. It is valuable because the institution of schooling converts birthdays into differences in required school attendance.

2.2 Section I: Season of birth, compulsory schooling, and years of education

Section I establishes the first-stage relationship. It asks whether quarter of birth actually affects schooling, and whether this effect is likely to be caused by compulsory schooling rather than unrelated seasonality.

- Figures I–III plot average schooling by quarter of birth for men born 1930–1959 using the 1980 Census.
- Figure IV detrends the schooling series and shows repeated downward deviations for early-year births.
- Table I estimates quarter-of-birth effects on educational outcomes.
- Table II uses variation in state school-leaving ages to show that compulsory laws affect enrollment around age 16.

Interpretation: This section must do more than show a correlation. It must show that the correlation has the right institutional source. The authors therefore combine graphical evidence, postsecondary placebo-style outcomes, and direct enrollment evidence.

2.3 Section I.A: Direct evidence on the effect of compulsory schooling laws

This subsection uses the fact that some states allow students to leave school at age 16, while other states require attendance until age 17 or 18. The key comparison is the enrollment rate of people who have already turned 16 versus people born later in the year who are still 15 at the Census date.

Interpretation: If compulsory schooling laws bind, then enrollment should fall more sharply at age 16 in states where the legal school-leaving age is 16. This is a difference-in-differences design: age/birthday differences are compared across law regimes.

2.4 Section I.B: Why do compulsory schooling laws work?

The paper discusses enforcement channels. Compulsory schooling laws are supported by child labor laws, work permit requirements, truant officers, and penalties on parents. There are exemptions, but the legal environment can still make leaving school costly for teenagers before the relevant dropout birthday.

Interpretation: This institutional discussion is important because the first stage is small. A small first stage is still informative if the law is enforceable and if the affected margin is the group of would-be early dropouts.

2.5 Section II: Estimating the return to education

Section II asks whether the quarter-of-birth schooling differences translate into earnings differences. It first presents Figure V and Table III, then estimates TSLS models.

- Figure V shows mean log weekly wages by quarter of birth for men born 1930–1949 in the 1980 Census.
- Table III reports Wald estimates: the reduced-form earnings difference divided by the first-stage schooling difference.
- Tables IV–VI report TSLS estimates for different cohorts.
- Table VII improves precision by allowing quarter-of-birth effects to vary by state of birth.
- Table VIII reports estimates for black men.

Interpretation: The section moves from intuitive IV estimates to fully controlled specifications. The core message is that the IV return to schooling is close to the OLS return.

2.6 Section II.A: TSLS estimates

This subsection formalizes the IV model. Education is predicted using quarter-of-birth indicators interacted with year-of-birth indicators. The wage equation includes education and controls. Quarter-of-birth interactions are excluded instruments.

Interpretation: Interacting quarter of birth with birth year lets the first-stage pattern vary across cohorts. This is important because the seasonal schooling pattern is not identical in every year.

2.7 Section II.B: Allowing the seasonal pattern in education to vary by state of birth

This subsection adds interactions between quarter-of-birth and state-of-birth dummies as instruments. State dummies are included in the wage equation, so identification comes from within-state seasonal variation in education.

Interpretation: This specification uses more institutional variation because school-entry policies vary across states and districts. It improves precision and strengthens the link between the instrument and the institutional mechanism.

2.8 Section II.C: Estimates for black men

This subsection estimates the same model for black men born 1930–1939. The paper expects lower returns for black men in these cohorts because school quality was likely lower.

Interpretation: This is a heterogeneity check. The estimates are lower than for all men, which the authors view as consistent with lower returns to compulsory schooling when school quality is lower.

2.9 Section III: Other possible effects of season of birth

Section III considers alternative reasons why season of birth might affect education or earnings.

- Older school entrants might be more mature or perform differently in school.
- Season of birth might be correlated with parental socioeconomic status.
- Quarter-of-birth dummies might affect earnings directly.
- Quarter of birth might matter for college graduates even though compulsory schooling should not bind for them.

The authors argue these alternatives do not explain the evidence. In particular, quarter of birth does not have the same pattern for postsecondary outcomes or for earnings of college graduates.

Interpretation: This section is the exclusion-restriction defense. The instrument is credible only if quarter of birth affects earnings through schooling rather than through other seasonal channels.

2.10 Section IV: Conclusion

The conclusion states that compulsory schooling laws increase educational attainment and that the extra schooling induced by the laws raises earnings. The similarity between OLS and TSLS estimates suggests little evidence of upward omitted-variable bias in conventional schooling regressions around the relevant schooling margin.

Interpretation: The paper does not prove that compulsory schooling is socially optimal. It estimates the private earnings effect of additional schooling induced by compulsory laws.

3 Data and measurement

3.1 Census data sets

The paper uses Public Use Census samples from 1960, 1970, and 1980.

- **1980 Census 5 percent sample:** used for the main quarter-of-birth education analysis for men born 1930–1949, and for earnings estimates for men born 1930–1949.
- **1970 Census:** used for earnings estimates for men born 1920–1929 and enrollment comparisons.
- **1960 Census 1 percent sample:** used for enrollment comparisons and school-entry-age evidence.
- **State samples:** used to classify compulsory attendance laws and to add state-of-birth interactions in the IV specifications.

Interpretation: The large Census samples are central. Quarter-of-birth differences are small, so the design requires very large samples to estimate first-stage and reduced-form effects precisely.

3.2 Main schooling outcomes

The paper studies several education outcomes:

- total years of completed schooling;
- high school graduation;
- years of education among high school graduates;
- college graduation;
- master’s degree completion;
- doctoral degree completion.

Interpretation: The postsecondary outcomes are important placebo-style outcomes. If quarter of birth mattered because of compulsory schooling, it should be strongest at the high-school dropout margin and much weaker for college and graduate-degree outcomes.

3.3 Earnings outcome

The main earnings outcome is log weekly wage:

$$\log W_i = \log \left(\frac{\text{annual wage and salary income}_i}{\text{weeks worked}_i} \right). \quad (1)$$

The main wage samples are restricted to men with positive wage/salary earnings and positive weeks worked.

Interpretation: Weekly wages focus on wage rates rather than labor supply. The paper also checks annual salary and weeks worked and argues that the main effect is on weekly wages, not on weeks worked.

3.4 Quarter of birth

Quarter of birth is coded as four mutually exclusive categories. Most regressions use dummies for the first three quarters, leaving the fourth quarter as the omitted category:

$$Q_{ij} = 1\{\text{individual } i \text{ was born in quarter } j\}, \quad j = 1, 2, 3. \quad (2)$$

Interpretation: The fourth quarter is a natural omitted group because children born late in the year are expected to start school younger and reach the dropout age later in their schooling careers.

3.5 Compulsory schooling law variables

The direct evidence in Table II classifies states by school-leaving age:

- states where students can leave at age 16;
- states where students must remain until age 17 or 18.

Interpretation: The law comparison is not the main IV strategy. It is supporting evidence that the legal mechanism actually binds.

3.6 Control variables

The TSLS wage regressions include different combinations of:

- year-of-birth dummies;
- age and age squared, measured to the quarter of a year;
- race dummies;
- SMSA residence;
- marital status;
- region-of-residence dummies;
- in Table VII, state-of-birth dummies.

Interpretation: Controls are included to absorb age trends, cohort differences, and broad demographic/geographic differences. Identification still comes from quarter-of-birth variation within birth years and, in Table VII, within states.

4 Empirical specifications and interpretation

4.1 OLS earnings equation

The conventional OLS model is:

$$\log W_i = X_i' \beta + \rho E_i + u_i, \quad (3)$$

where E_i is years of education and X_i includes cohort and demographic controls.

Interpretation: OLS estimates can be biased if schooling is correlated with ability, family background, measurement error, or other unobserved determinants of earnings.

4.2 Quarter-of-birth first stage

A simplified first-stage equation is:

$$E_i = X_i' \pi + \sum_c \gamma_c Y_{ic} + \sum_{j=1}^3 \sum_c \theta_{jc} (Q_{ij} Y_{ic}) + v_i, \quad (4)$$

where Y_{ic} are year-of-birth dummies.

Interpretation: The coefficients on quarter-by-year interactions measure the schooling differences across quarters of birth within the same birth cohort. These are the excluded instruments.

4.3 TSLS wage equation

The second stage is:

$$\log W_i = X_i' \beta + \sum_c \delta_c Y_{ic} + \rho E_i + u_i. \quad (5)$$

Education is instrumented with the quarter-of-birth by year-of-birth interactions.

Interpretation: The coefficient ρ is interpreted as the return to an additional year of schooling for people whose schooling is shifted by compulsory attendance laws.

4.4 Wald estimator

Table III uses a simple Wald estimator comparing first-quarter births with births in the other three quarters:

$$\hat{\rho}_{Wald} = \frac{\overline{\log W}_{Q1} - \overline{\log W}_{Q2-Q4}}{\overline{E}_{Q1} - \overline{E}_{Q2-Q4}}. \quad (6)$$

Interpretation: This is the simplest IV estimator. It is the reduced-form earnings effect divided by the first-stage schooling effect.

4.5 State-by-quarter instruments

Table VII adds state variation:

$$E_i = X_i' \pi + \sum_c \gamma_c Y_{ic} + \sum_s \lambda_s S_{is} + \sum_{j,c} \theta_{jc} (Q_{ij} Y_{ic}) + \sum_{j,s} \kappa_{js} (Q_{ij} S_{is}) + v_i. \quad (7)$$

The wage equation includes state-of-birth fixed effects, so the extra instruments are excluded quarter-by-state interactions.

Interpretation: This specification recognizes that the mapping from birthday to school-entry age varies geographically. The paper reports that allowing state-specific seasonality reduces IV standard errors.

4.6 Difference-in-differences enrollment calculation

Table II estimates whether 16-year-old enrollment falls more in states where age 16 is the legal school-leaving age. The basic contrast is:

$$\Delta\Delta = [Enroll_{Q1,Law16} - Enroll_{Q2-Q4,Law16}] - [Enroll_{Q1,Law17/18} - Enroll_{Q2-Q4,Law17/18}]. \quad (8)$$

Interpretation: The within-state-law comparison removes broad age effects. The cross-law comparison asks whether the birthday effect is stronger where students are legally allowed to leave at age 16.

4.7 Overidentification test

Because the TSLS models use many quarter-by-year instruments, the paper tests whether the different instruments imply the same return to schooling. The statistic is based on the sample size times the R^2 from regressing TSLS residuals on the exogenous variables and instruments.

Interpretation: Passing the overidentification test supports the idea that the instruments are all capturing the same schooling-induced earnings effect. Rejections would suggest heterogeneity, invalid instruments, or finite-sample problems.

5 Identification and credibility

5.1 Relevance

Quarter of birth is relevant because school-entry rules and dropout-age laws create different minimum schooling lengths.

- First-quarter births tend to enter school older.
- They reach age 16 or 17 earlier in their school careers.
- If they want to drop out, they can legally do so with less completed schooling.
- Later-year births are forced to remain enrolled longer before reaching the same legal age.

Interpretation: This explains why the first stage should be strongest near the high-school dropout margin, not in college completion.

5.2 Exclusion restriction

The exclusion restriction is:

$$Cov(Q_{ij}, u_i) = 0 \quad \text{conditional on controls,} \quad (9)$$

meaning quarter of birth affects earnings only through schooling.

Interpretation: The paper spends substantial effort defending this assumption. It is plausible only if season of birth is not correlated with ability, parental background, health, or labor-market opportunities in a way that independently changes earnings.

5.3 Age effects

Quarter of birth is mechanically correlated with age within a Census year. This matters because earnings rise with age for younger workers.

- The paper emphasizes men aged 40–49 because their age-earnings profile is relatively flat.
- It includes age and age-squared controls in some specifications.
- It finds that simple Wald estimates for younger men can be misleading because the age-earnings profile is still upward sloping.

Interpretation: This is a crucial design choice. If older first-quarter workers earn more simply because they are older, the reduced form could be biased upward. The paper therefore focuses on ages where this channel is weak.

5.4 Seasonality and family background

A concern is that births in different seasons may reflect parental socioeconomic status. The paper cites evidence suggesting seasonality of births is not strongly tied to socioeconomic status and argues that such explanations do not naturally explain why the schooling effect is weaker in more recent cohorts.

Interpretation: The authors' defense is not that seasonality is impossible. It is that the specific schooling patterns are more consistent with compulsory-schooling constraints than with general seasonal selection.

5.5 Postsecondary and college-graduate checks

The paper tests whether quarter of birth affects educational outcomes beyond the compulsory-schooling margin and earnings among college graduates.

- Table I shows no clear systematic quarter-of-birth pattern for graduate-degree completion.
- The paper reports that quarter of birth has no meaningful effect on earnings for college graduates.

Interpretation: These checks are powerful because college graduates are not constrained by minimum school-leaving laws. If quarter of birth predicted their earnings, the exclusion restriction would be less credible.

5.6 LATE interpretation

The IV estimate applies to compliers: students who would have left school earlier but are compelled to stay longer because of the legal school-leaving age.

Interpretation: The estimate is not necessarily the average return to schooling for all workers. It is the return for marginal dropouts induced to acquire more schooling by compulsory laws.

5.7 Limitations and later econometric issues

- The first stage is small, so precision depends on enormous samples.
- Specifications with many instruments can raise finite-sample concerns.
- The paper predates much of the modern weak-instrument and many-instrument literature.

- Quarter-of-birth instruments identify a narrow local effect, not necessarily returns to college or higher education.
- The design estimates private earnings gains, not the full social costs and benefits of compulsory schooling.

Interpretation: These limitations do not overturn the paper’s contribution, but they matter for how the estimates should be used.

6 Tables and figures

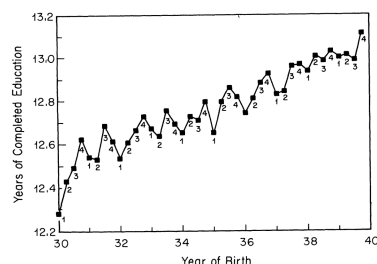
This section preserves the original visuals by directly cropping them from the paper. Adjacent figures and tables are grouped to save space and make comparisons faster.

6.1 Figures I–III: schooling by quarter of birth across cohorts

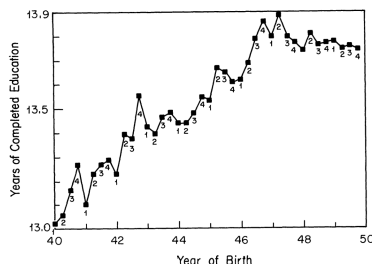
Read:

- Figure I covers men born in the 1930s; Figure II covers men born in the 1940s; Figure III covers men born in the 1950s.
- The quarter number is plotted under each observation.
- Across many birth years, men born in the first quarter have slightly lower completed schooling than men born later in the year.
- The pattern is small, but highly persistent and visible across many cohorts.
- The effect appears weaker for later cohorts, consistent with compulsory schooling becoming less binding as average schooling rises.

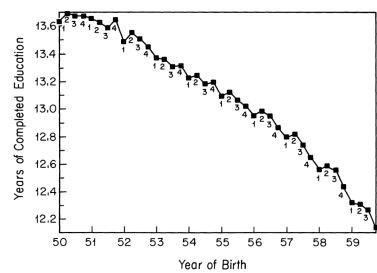
Economic message: The figures visually establish the first stage. Quarter of birth is not a large determinant of schooling, but it creates systematic variation at the dropout margin.



(a) Figure I



(b) Figure II



(c) Figure III

Years of education and season of birth, directly cropped from the paper.

6.2 Figure IV and Table I: detrended schooling and education outcomes

Read: Figure IV:

- Figure IV detrends schooling using a moving-average transformation and plots deviations from trend.
- The first-quarter observations repeatedly show negative deviations.

- The figure makes the seasonal first-stage pattern easier to see because it removes cohort-level schooling trends.

Read: Table I:

- For total years of education, first-quarter births have 0.124 fewer years in the 1930–1939 cohort and 0.085 fewer years in the 1940–1949 cohort, relative to fourth-quarter births.
- For high school graduation, first-quarter births are 1.9 percentage points lower in the 1930s cohort and 1.5 percentage points lower in the 1940s cohort.
- Quarter-of-birth effects are much weaker and less systematic for postsecondary outcomes.
- Master’s and doctoral degree outcomes do not show the same clear compulsory-schooling pattern.

Economic message: The compulsory-schooling mechanism predicts effects at the high-school dropout margin, not uniformly across all education outcomes. Table I matches that prediction.

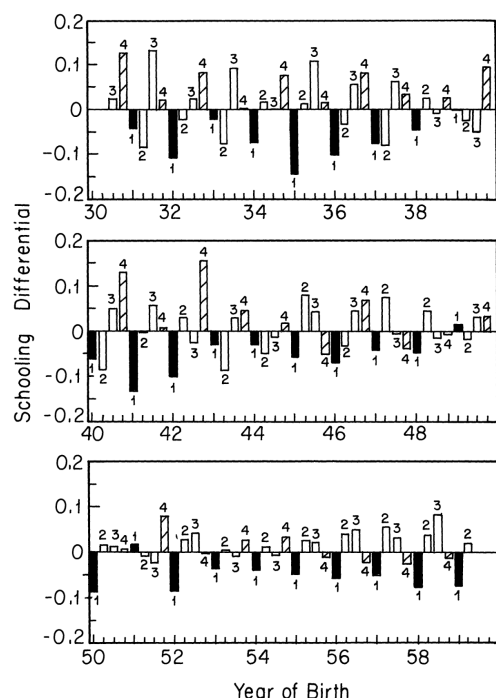


FIGURE IV
Season of Birth and Years of Schooling
Deviations from $MA(+2, -2)$

(a) Figure IV

TABLE I
THE EFFECT OF QUARTER OF BIRTH ON VARIOUS EDUCATIONAL
OUTCOME VARIABLES

Outcome variable	Birth cohort	Mean	Quarter-of-birth effect ^a			<i>F</i> -test ^b [<i>P</i> -value]
			I	II	III	
Total years of education	1930–1939	12.79	-0.124 (0.017)	-0.086 (0.017)	-0.015 (0.016)	24.9 [0.0001]
	1940–1949	13.56	-0.085 (0.012)	-0.035 (0.012)	-0.017 (0.011)	18.6 [0.0001]
High school graduate	1930–1939	0.77	-0.019 (0.002)	-0.020 (0.002)	-0.004 (0.002)	46.4 [0.0001]
	1940–1949	0.86	-0.015 (0.001)	-0.012 (0.001)	-0.002 (0.001)	54.4 [0.0001]
Years of educ. for high school graduates	1930–1939	13.99	-0.004 (0.014)	0.051 (0.014)	0.012 (0.014)	5.9 [0.0006]
	1940–1949	14.28	0.005 (0.011)	0.043 (0.011)	-0.003 (0.010)	7.8 [0.0017]
College graduate	1930–1939	0.24	-0.005 (0.002)	0.003 (0.002)	0.002 (0.002)	5.0 [0.0021]
	1940–1949	0.30	-0.003 (0.002)	0.004 (0.002)	0.000 (0.002)	5.0 [0.0018]
Completed master’s degree	1930–1939	0.09	-0.001 (0.001)	0.002 (0.001)	-0.001 (0.001)	1.7 [0.1599]
	1940–1949	0.11	0.000 (0.001)	0.004 (0.001)	0.001 (0.001)	3.9 [0.0091]
Completed doctoral degree	1930–1939	0.03	0.002 (0.001)	0.003 (0.001)	0.000 (0.001)	2.9 [0.0332]
	1940–1949	0.04	-0.002 (0.001)	0.001 (0.001)	-0.001 (0.001)	4.3 [0.0050]

a. Standard errors are in parentheses. An $MA(+2, -2)$ trend term was subtracted from each dependent variable. The data set contains men from the 1980 Census, 5 percent Public Use Sample. Sample size is 312,718 for 1930–1939 cohort and is 457,181 for 1940–1949 cohort.

b. *F*-statistic is for a test of the hypothesis that the quarter-of-birth dummies jointly have no effect.

(b) Table I

Quarter-of-birth effects on schooling and educational outcomes.

6.3 Table II: direct evidence from state dropout-age laws

Read:

- Table II compares age-16 enrollment in states with school-leaving age 16 versus states with school-leaving age 17 or 18.

- In 1960, the within-law enrollment decline from age 15 to age 16 is much larger in age-16 states than in age-17/18 states; the difference-in-differences is about -4.0 percentage points.
- In 1970, the analogous difference-in-differences is about -2.0 percentage points.
- By 1980, the effect is smaller, consistent with compulsory schooling becoming less binding as high-school completion becomes more common.

Economic message: This table validates the legal mechanism. Some students leave school when they are legally allowed to leave, so compulsory attendance laws do increase schooling for marginal students.

TABLE II
PERCENTAGE OF AGE GROUP ENROLLED IN SCHOOL BY BIRTHDAY AND LEGAL
DROPOUT AGE^a

Date of birth	Type of state law ^b		Column (1) – (2)
	School-leaving age: 16 (1)	School-leaving age: 17 or 18 (2)	
Percent enrolled April 1, 1960			
1. Jan 1–Mar 31, 1944 (age 16)	87.6 (0.6)	91.0 (0.9)	–3.4 (1.1)
2. Apr 1–Dec 31, 1944 (age 15)	92.1 (0.3)	91.6 (0.5)	0.5 (0.6)
3. Within-state diff. (row 1 – row 2)	–4.5 (0.7)	–0.6 (1.0)	–4.0 (1.2)
Percent enrolled April 1, 1970			
4. Jan 1–Mar 31, 1954 (age 16)	94.2 (0.3)	95.8 (0.5)	–1.6 (0.6)
5. Apr 1–Dec 31, 1954 (age 15)	96.1 (0.1)	95.7 (0.3)	0.4 (0.3)
6. Within-state diff. (row 1 – row 2)	–1.9 (0.3)	0.1 (0.6)	–2.0 (0.6)
Percent enrolled April 1, 1980			
7. Jan 1–Mar 31, 1964 (age 16)	95.0 (0.1)	96.2 (0.2)	–1.2 (0.2)
8. Apr 1–Dec 31, 1964 (age 15)	97.0 (0.1)	97.7 (0.1)	–0.7 (0.1)
9. Within-state diff. (row 1 – row 2)	–2.0 (0.1)	–1.5 (0.2)	0.5 (0.3)

a. Standard errors are in parentheses.

b. Data set used to compute rows 1–3 is the 1960 Census, 1 percent Public Use Sample; data set used to compute rows 4–6 is 1970 Census, 1 percent State Public Use Sample (15 percent form); data set used to compute rows 7–9 is the 1980 Census, 5 percent Public Use Sample. Each sample contains both boys and girls.

Table II: Enrollment by birthday and legal dropout age.

6.4 Figure V and Table III: reduced-form earnings and Wald IV estimates

Read: Figure V:

- Figure V plots mean log weekly wages by quarter of birth for men born 1930–1949 in the 1980 Census.
- First-quarter births tend to earn slightly less, consistent with their lower schooling.
- The age-earnings profile is upward sloping for younger men but flatter for 40–49-year-old men, motivating the paper’s focus on the older group.

Read: Table III:

- For 1970 men born 1920–1929, first-quarter births earn 0.00898 log points less and have 0.1256 fewer years of education; the Wald return is 0.0715.
- For 1980 men born 1930–1939, first-quarter births earn 0.01110 log points less and have 0.1088 fewer years of education; the Wald return is 0.1020.

- The corresponding OLS returns are 0.0801 and 0.0709.
- Wald and OLS estimates are statistically close.

Economic message: The simple IV ratio already contains the main message: the earnings gain from compulsory schooling is similar to conventional OLS estimates.

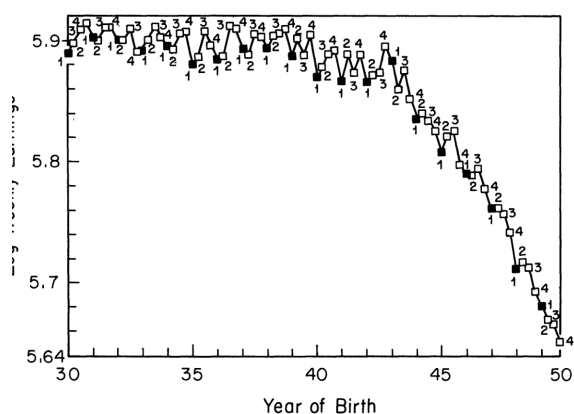


FIGURE V
Mean Log Weekly Wage, by Quarter of Birth
All Men Born 1930–1949; 1980 Census

(a) Figure V

	(1) Born in 1st quarter of year	(2) Born in 2nd, 3rd, or 4th quarter of year	(3) Difference (std. error) (1) – (2)
ln (wkly. wage)	5.1484	5.1574	–0.00898 (0.00301)
Education	11.3996	11.5252	–0.1256 (0.0155)
Wald est. of return to education			0.0715 (0.0219)
OLS return to education ^b			0.0801 (0.0004)

	(1) Born in 1st quarter of year	(2) Born in 2nd, 3rd, or 4th quarter of year	(3) Difference (std. error) (1) – (2)
ln (wkly. wage)	5.8916	5.9027	–0.01110 (0.00274)
Education	12.6881	12.7969	–0.1088 (0.0132)
Wald est. of return to education			0.1020 (0.0239)
OLS return to education			0.0709 (0.0003)

a. The sample size is 247,199 in Panel A, and 327,509 in Panel B. Each sample consists of males born in the United States who had positive earnings in the year preceding the survey. The 1980 Census sample is drawn from the 5 percent sample, and the 1970 Census sample is from the State, County, and Neighborhoods 1 percent samples.
b. The OLS return to education was estimated from a bivariate regression of log weekly earnings on years of education.

(b) Table III

Reduced-form wage pattern and Wald estimates.

6.5 Tables IV–VI: baseline OLS and TSLS estimates by cohort

Read: Table IV:

- Table IV uses 1970 Census men aged 40–49, born 1920–1929.
- The OLS return is about 0.080 in the simplest specification.
- The comparable TSLS estimate is about 0.077.
- Adding age controls and demographic/geographic controls does not create a large gap between OLS and TSLS.

Read: Table V:

- Table V uses 1980 Census men aged 40–49, born 1930–1939.
- In the richer specifications, the OLS return is about 0.063.
- The comparable TSLS estimate is about 0.060 in the baseline quarter-by-year instrument set.
- Again, OLS and TSLS estimates are very close.

- The authors interpret this as consistent with lower school quality for black students in these cohorts.
- TSLS estimates are slightly below OLS estimates in this subsample, but the differences are within sampling variation.

Economic message: State-specific instruments improve precision, while the black-men subsample offers a substantive heterogeneity check related to school quality.

TABLE VIII
OLS AND TSLS ESTIMATES OF THE RETURN TO EDUCATION FOR BLACK MEN BORN 1930–1939: 1980 CENSUS^a

Independent variable	(1) OLS	(2) TSLS	(3) OLS	(4) TSLS	(5) OLS	(6) TSLS	(7) OLS	(8) TSLS
Years of education	0.0672 (0.0013)	0.0635 (0.0185)	0.0671 (0.0003)	0.0533 (0.0199)	0.0576 (0.0013)	0.0461 (0.0187)	0.0576 (0.0013)	0.0391 (0.0199)
SMSA (1 = center city)	—	—	—	—	0.1885 (0.0142)	0.2053 (0.0308)	0.1884 (0.0142)	0.2155 (0.0324)
Married (1 = married)	—	—	—	—	0.2216 (0.0193)	0.2272 (0.0136)	0.2216 (0.0100)	0.2307 (0.0140)
9 Year-of-birth dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
8 Region-of-residence dummies	No	No	No	No	Yes	Yes	Yes	Yes
49 State-of-birth dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Age	—	—	—0.0309 (0.2538)	—0.3274 (0.2560)	—	—	—0.2978 (0.0032)	—0.3237 (0.2497)
Age-squared	—	—	0.0033 (0.0028)	0.0035 (0.0028)	—	—	0.0032 (0.0027)	0.0035 (0.0028)
χ^2 (dof)	—	184 [176]	—	181 [173]	—	178 [176]	—	175 [173]

^a Standard errors are in parentheses. Excluded instruments are 36 quarter-of-birth times year-of-birth dummies and 147 quarter-of-birth times state-of-birth interactions. (There are no black men in the sample born in Hawaii.) Age and age-squared are measured in quarters of years. Each equation also includes an intercept term. The sample is drawn from the 1980 Census. Sample size is 26,913.

Table VIII: estimates for black men born 1930–1939.

6.7 Appendix 2: compulsory school attendance age by state

Read:

- Appendix 2 lists compulsory school attendance ages by state in 1960, 1970, and 1980.
- Most states require schooling until age 16; some require attendance until 17 or 18.
- Several states changed rules or had special exemptions.
- This institutional variation supports Table II and the state-interaction strategy.

Economic message: The appendix shows that the legal variation used in the paper is real and state-specific rather than a purely conceptual argument.

APPENDIX 2: (CONTINUED)

APPENDIX 2: COMPULSORY SCHOOL ATTENDANCE AGE BY STATE

State	1960	1970	1980	Notes
1 Alabama	16	16	16	
2 Alaska	16	16	16	
4 Arizona	16	16	16	
5 Arkansas	16	16	15	
6 California	16	16	16	
8 Colorado	16	16	16	
9 Connecticut	16	16	16	
10 Delaware	16	16	16	
11 D.C.	16	16	16	
12 Florida	16	16	16	
13 Georgia	16	16	16	
15 Hawaii	16	18	18	Increased to 18 midyear, 1970
16 Idaho	16	16	16	
17 Illinois	16	16	16	

(a) Appendix 2, part 1

State	1960	1970	1980	Notes
18 Indiana	16	16	16	
19 Iowa	16	16	16	
20 Kansas	16	16	16	
21 Kentucky	16	16	16	
22 Louisiana	16	16	16	
23 Maine	15	17	17	
24 Maryland	16	16	16	
25 Massachusetts	16	16	16	
26 Michigan	16	16	16	
27 Minnesota	16	16	16	
28 Mississippi	—	—	—	Age 14 starting 1983
29 Missouri	16	16	16	
30 Montana	16	16	16	
31 Nebraska	16	16	16	
32 Nevada	17	17	17	
33 New Hampshire	16	16	16	
34 New Jersey	16	16	16	
35 New Mexico	16	17	17	
36 New York	16	16	16	May be changed by city districts
37 North Carolina	16	16	16	
38 North Dakota	17	16	16	
39 Ohio	18	18	18	
40 Oklahoma	18	18	18	
41 Oregon	18	18	18	
42 Pennsylvania	17	17	17	
44 Rhode Island	16	16	16	
45 South Carolina	—	16	16	Reinstated in 1967
46 South Dakota	16	16	16	
47 Tennessee	17	17	16	Increased to age 17 in 1983
48 Texas	16	17	17	Must finish school term
49 Utah	18	18	18	
50 Vermont	16	16	16	
51 Virginia	16	17	17	
53 Washington	16	16	18	
54 West Virginia	16	16	16	
55 Wisconsin	16	16	16	Must finish school term
54 Wyoming	17	17	16	

Source: U.S. Office of Education, *Digest of Education Statistics* (Washington, DC: GPO, various years).

(b) Appendix 2, part 2

Compulsory school attendance age by state.

7 Interpretation by major paper section

7.1 Why quarter of birth changes schooling

The mechanism is mechanical but subtle. Suppose two students enter school in the same academic cohort. A child born early in the calendar year is older at school entry. When both students approach age 16, the early-born child reaches the legal dropout age earlier in the academic sequence. If both children want to leave school as soon as legally allowed, the early-born child exits with less completed schooling.

The empirical first stage therefore requires three ingredients:

1. school entry is tied to a cutoff date;
2. dropout permission is tied to age, not grade completion;
3. some students would like to leave school before graduating.

Interpretation: This is why the effect is concentrated at the lower tail of schooling. The design does not imply that quarter of birth strongly affects high-achieving students.

7.2 Why the first stage is small but meaningful

The effect of being born in the first quarter is less than one-tenth of a year of education for the 1940s cohort and about one-eighth of a year for the 1930s cohort. This is small because only potential dropouts are affected. Students who would graduate high school or attend college regardless of the law are never shifted by the instrument.

Interpretation: A small first stage does not mean the law is unimportant. It means the law affects a small group: students near the legal dropout margin.

7.3 Why the OLS–IV comparison is economically important

The conventional concern is that high-ability people get more schooling and earn more, causing OLS returns to schooling to be biased upward. The paper’s IV estimates are not below OLS. This suggests either:

- ability bias in OLS is small at this schooling margin;
- measurement error biases OLS downward and offsets ability bias;
- the marginal return for compulsory-schooling compliers is higher than the average OLS return;
- or some combination of these forces.

Interpretation: The paper does not prove that all OLS schooling regressions are unbiased. It shows that for men around the compulsory-schooling margin, IV and OLS are surprisingly similar.

7.4 Why college graduates are a falsification group

College graduates are not forced to complete college by compulsory schooling laws. If quarter of birth affected earnings through seasonality, health, maturity, or family background, it should also show up in college-graduate earnings. The absence of such a pattern supports the exclusion restriction.

Interpretation: This is an early and intuitive example of a placebo test in quasi-experimental research.

7.5 Why the state-interaction specification matters

State school-entry policies and attendance laws vary. If one state admits children by a different cutoff rule, quarter of birth may have a different effect on completed schooling. By interacting quarter of birth with state of birth, the paper extracts more variation from the institutional environment.

Interpretation: This is both a precision improvement and a credibility improvement. The instrument becomes more closely tied to the legal-schooling mechanism.

7.6 Why the black-men estimates are lower

The paper argues that black men in the earlier cohorts often attended lower-quality schools. If the return to a forced extra year of schooling depends on school quality, then the return should be lower for black men in these cohorts. Table VIII is consistent with this prediction.

Interpretation: This is not the main estimate. It is evidence that the IV strategy produces economically sensible heterogeneity.

7.7 What the paper says about compulsory schooling policy

The paper estimates a private earnings effect. It does not conduct a full welfare analysis. Compulsory schooling may also affect crime, civic outcomes, peer effects, school discipline, public spending, and the learning environment of other students.

Interpretation: A positive private return supports the idea that compulsory schooling benefits the marginal student, but it does not by itself prove that the law is socially optimal.

8 Short summary

- The paper exploits quarter of birth as an instrument for schooling.
- The institutional mechanism is the interaction of school-entry cutoffs and compulsory attendance laws.
- First-quarter births start school older and can leave school after completing less schooling.
- Figures I–IV and Table I show that first-quarter births have slightly less education.
- Table II shows that enrollment falls more at age 16 in states where age 16 is the legal school-leaving age.
- Figure V shows that first-quarter births earn slightly less.
- Table III's Wald estimates are similar to OLS estimates.
- Tables IV–VI show that TSLS and OLS estimates are close across cohorts and controls.
- Table VII adds state-specific quarter-of-birth instruments and improves precision.
- Table VIII shows lower returns for black men, consistent with differences in school quality.
- Section III defends the exclusion restriction against age-at-entry, family-background, and direct seasonality concerns.
- The main conclusion is that compulsory schooling increases education and that the induced schooling raises earnings.
- The similarity between IV and OLS suggests little upward omitted-variable bias in OLS returns to schooling around the compulsory-schooling margin.

9 Conclusion

9.1 Core conclusion

Angrist and Krueger show that compulsory schooling laws affect educational attainment and that the additional schooling induced by these laws raises earnings. Their IV estimates of the return to schooling are close to conventional OLS estimates, suggesting that OLS returns to schooling are not strongly upward biased for the marginal compulsory-schooling population.

9.2 Economic intuition

The economic intuition is that legal rules create an accidental difference in required schooling. Two students with similar preferences and ability but different birthdays can face different minimum schooling requirements. The labor-market return to this extra legally induced schooling is then estimated by comparing the earnings difference across birthdays to the schooling difference across birthdays.

9.3 Methodological takeaway

The paper is a landmark because it turns an institutional detail—quarter of birth relative to school-entry and dropout-age rules—into a source of identification. It illustrates the logic of instrumental variables: find variation in treatment that is plausibly unrelated to unobserved determinants of the outcome, verify that it shifts the treatment, and interpret the ratio of reduced form to first stage.

9.4 Limits of the result

The estimate is local to students affected by compulsory schooling. It should not be automatically generalized to college, graduate school, or all education margins. The first stage is small, and specifications with many instruments raise concerns that later econometric work explored. Still, the paper remains a foundational example of quasi-experimental research in labor economics.